Robotic Arm

BY GONE RAMU

Project Aim

- To control a 4 DOF robotic arm by mobile using HC – 05 Bluetooth Module

Apparatus Required

Hardware Tools :

1. Arduino Mega
2. Bluetooth HC – 05 module
3. 4 Servo Motors
4. Jumper wires
5. Arduino cable
6. Breadboard
7. Battery Source
8. Screws and nuts

Apparatus Required

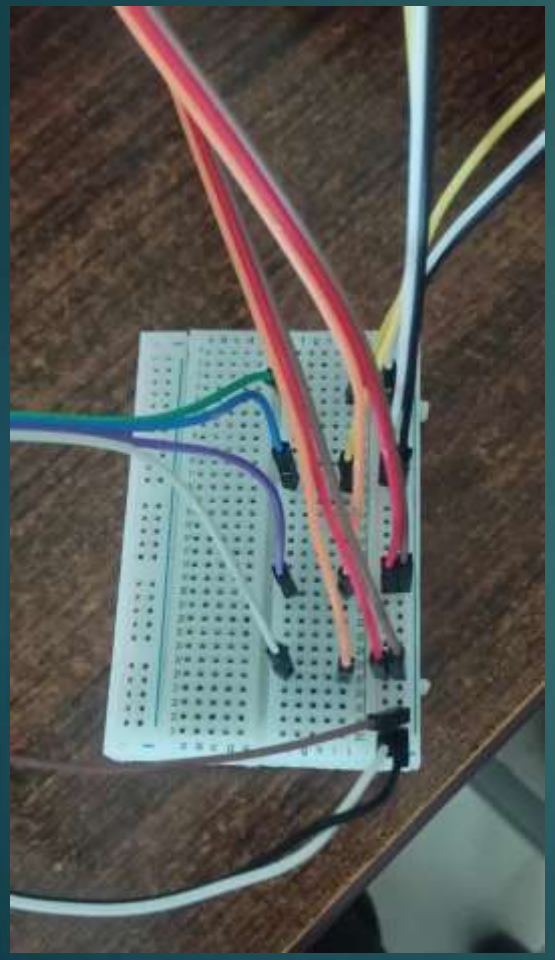
Software Tools :

1. Arduino IDE
2. UI Interface
3. Arduino code of Robotic Arm
4. Arduino code of Bluetooth Controlled Robotic Arm
5. Robotic Arm cad file

Procedure

- Arrange 4 Servos in according to Robotic arm structure along with its parts.
- Integrate 4 Servos with Arduino mega.
- Now connect the Bluetooth HC – 05 module to Arduino mega.
- Connect Batteries to Servos for supplying the power.
- Open the Arduino IDE
- Upload the Robotic Arm Code
- Turn on the Bluetooth in your mobile and open UI application.
- Click on connect. Click on “HC – 05 Bluetooth module”
- Bluetooth connection is Done.

Circuit Connection





Servos connection with Batteries :

4 Servos	Battery
VCC	Anode
GND	Cathode/GND

Servos connection with Arduino mega :

Servo signal pin	Arduino mega
Servox	9
Servoy	10
Servoz	11
Servog	12

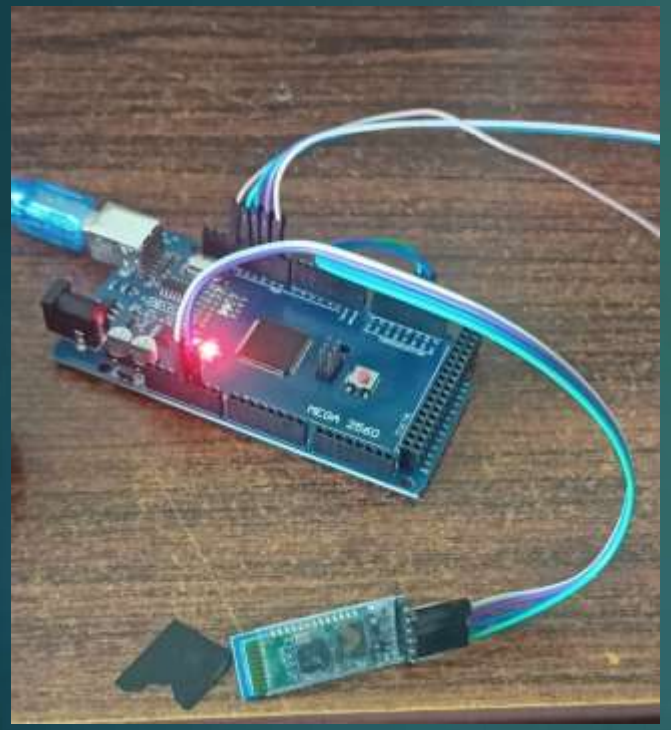
Circuit Connection



Integration of Bluetooth module with Arduino mega :

Bluetooth module	Arduino mega
TX	RX (19)
RX	TX (18)
VCC	5v
GND	GND

UI Application Connection









Before Bluetooth connection

After Bluetooth connection

Result

- Finally, we controlled 4 – DOF Robotic Arm by mobile UI interface using Bluetooth module.